

Computer Networking

Course Syllabus & Session Plan

Course Type	Skill Development	Course Category	Discipline Specific Course (DSC)
Theory Hours	23	Practical Hours	7
Guided Hours	15	Total Hours / Credits	30 Hours / 2 Credits

Course Description

This course provides a comprehensive foundation in computer networking, covering core concepts, LAN switching, IP routing, and network security. Learners begin with networking fundamentals such as topologies, the OSI/TCP-IP models, addressing, and transmission media. The course then moves into LAN switching technologies including VLANs, trunking, and Spanning Tree Protocol, followed by an in-depth study of static and dynamic routing (OSPF, EIGRP, and BGP), access control lists, and NAT. The course concludes with an introduction to network security, covering common threats, firewalls, VPNs, and hardening practices. Throughout the course, learners also build the analytical skill to compare protocol behavior and troubleshoot switching, routing, and security issues, reinforced through hands-on labs using simulation tools such as Cisco Packet Tracer/GNS3 and Wireshark.

Course Objective

To build a strong conceptual and practical foundation in computer networking, enabling learners to describe core concepts, design and configure switched and routed networks, analyze and troubleshoot network behavior, and apply network security practices to protect network infrastructure.

Course Outcomes (COs)

CO	Description	Cognition	Hours	Evaluation
CO1	Describe fundamental networking concepts, the OSI/TCP-IP models, IP addressing, and transmission media.	L2: Understand	6	Continuous Assignment
CO2	Configure LAN switching technologies including VLANs, trunking, and Spanning Tree Protocol.	L3: Apply	8	Continuous Test
CO3	Configure static and dynamic routing protocols (OSPF, EIGRP, BGP) along with ACLs and NAT.	L3: Apply	8	Continuous Project
CO4	Apply network security fundamentals including firewalls, VPNs, and hardening techniques to protect network infrastructure.	L3: Apply	4	Continuous Case Study
CO5	Analyze and troubleshoot switching and routing behavior by comparing protocol performance (STP, OSPF, EIGRP, BGP), diagnosing connectivity faults, and examining network traffic using diagnostic tools such as Wireshark.	L4: Analyze	4	Continuous Lab/Diagnostic Assessment

Syllabus

Learning Objectives	CO	L/T/P	Module	Sub-Topics	Prescribed Texts	Online References
Understand fundamental networking concepts, models, addressing, and media types.	CO1	T	Module I: Networking Basics	Network types & topologies; OSI & TCP/IP models; networking devices; transmission media; IP addressing & subnetting; core protocols (ARP, ICMP, DHCP, DNS).	Forouzan — Data Communications and Networking; Tanenbaum — Computer Networks	geeksforgeeks.org/computer-network-tutorials
Configure and troubleshoot LAN switching technologies.	CO2	T/P	Module II: LAN Switching	Switch operation & switching methods; VLANs & trunking; inter-VLAN routing; STP/RSTP; EtherChannel; switch port security; VTP.	Wendell Odom — CCNA 200-301 Official Cert Guide, Vol. 1	ciscopress.com
Configure and troubleshoot static and dynamic IP routing.	CO3	T/P	Module III: Routing	Static routing; dynamic routing overview; OSPF; EIGRP; BGP fundamentals & attributes; ACLs; NAT.	Wendell Odom — CCNA 200-301 Official Cert Guide, Vol. 2; Jeff Doyle — Routing TCP/IP	practicalnetworking.net
Apply fundamental network security and hardening practices.	CO4	T/P	Module IV: Network Security	CIA triad & common threats; sniffing, spoofing, MITM, DoS/DDoS; firewalls & VPNs; device hardening; wireless security (WPA2/WPA3).	William Stallings — Network Security Essentials	cisco.com/c/en/us/products/security
Analyze and troubleshoot switching and routing behavior using performance comparison and diagnostic techniques.	CO5	P	Module II & III: Switching & Routing Analysis	STP/RSTP variant comparison & convergence troubleshooting; switching fault diagnosis; BGP route verification & troubleshooting; static/OSPF/EIGRP/BGP routing troubleshooting scenarios; packet-level traffic analysis with Wireshark.	Wendell Odom — CCNA 200-301 Official Cert Guide, Vol. 1 & 2; Jeff Doyle — Routing TCP/IP	wireshark.org/docs

Session Plan (30 Hours)

#	Session Topic	CO	L/T/P	Sub-Topics	Pedagogy	Assessment
1	Introduction to Computer Networks & Network Topologies	CO1	T	What is a network, need for networking, types (LAN, MAN, WAN, PAN); Bus, star, ring, mesh, hybrid topologies, advantages/disadvantages	Lecture + Discussion	-
2	OSI Reference Model & TCP/IP Model	CO1	T	7 layers, functions of each layer, encapsulation/decapsulation; 4/5-layer model, comparison with OSI, protocol suite overview	Lecture + Discussion	-
3	Networking Devices & Transmission Media	CO1	T	Hub, switch, router, bridge, repeater, gateway, access point; Guided media (UTP, STP, coaxial, fiber) and unguided media (wireless, satellite)	Lecture + Discussion	-
4	IP Addressing Fundamentals & Subnetting Basics	CO1	T	IPv4 structure, classes (A-E), public vs private IP, special addresses; Subnet mask, CIDR notation, calculating subnets and hosts	Lecture + Discussion	-
5	Introduction to Protocols	CO1	T	Ethernet, ARP, ICMP, DHCP, DNS - roles and basic working	Lecture + Discussion	-
6	Basics Review & Hands-on Lab	CO1	P	Cabling practice, basic ping/traceroute exercises, IP configuration, quiz	Lab Demonstration + Hands-on Practice	Quiz 1 + Lab Evaluation (Cabling & IP Configuration)
7	Introduction to LAN Switching & Switch Operation	CO2	T	Role of switches, collision domains, broadcast domains; MAC address learning, forwarding, filtering, flooding	Lecture + Discussion	-
8	Switching Methods & Ethernet Frame Structure	CO2	T	Store-and-forward, cut-through, fragment-free switching; Frame format, addressing, MTU	Lecture + Discussion	-
9	VLAN Fundamentals & VLAN Configuration	CO2	T	What is a VLAN, benefits, VLAN types; Creating VLANs, assigning ports, VLAN database	Lecture + Discussion	-
10	Trunking & Inter-VLAN Routing	CO2	T	802.1Q tagging, native VLAN, trunk port configuration; Router-on-a-stick, Layer 3 switching concepts	Lecture + Discussion	-
11	Spanning Tree Protocol (STP) & STP Variants	CO5	T	Loop prevention, root bridge election, port states; RSTP, PVST+, comparison and use cases	Lecture + Discussion	-
12	EtherChannel / Link Aggregation	CO2	T	Purpose, configuration basics, load balancing	Lecture + Discussion	-
13	Switch Security Basics	CO2	T	Port security, MAC filtering, DHCP snooping overview	Lecture + Discussion	-
14	VTP (VLAN Trunking Protocol)	CO2	T	Modes (server, client, transparent), synchronization	Lecture + Discussion	-
15	Switching Lab Session 1	CO2	P	Configure VLANs, trunks, and verify connectivity	Lab Demonstration + Hands-on Practice	Lab Evaluation (VLAN & Trunk Configuration)
16	Switching Lab Session 2 & Review	CO5	P	STP verification, troubleshooting switching issues, quiz	Lab Demonstration + Hands-on Practice	Quiz 2 + Lab Evaluation (STP)
17	Introduction to Routing & Static Routing	CO3	T	Role of routers, routing table, path determination; Configuring static routes, default routes, use cases	Lecture + Discussion	-
18	Dynamic Routing Overview & Routing Metrics / Administrative Distance	CO3	T	Need for dynamic routing, classification (distance-vector, link-state, path-vector); How routers choose the best path	Lecture + Discussion	-
19	OSPF Fundamentals & OSPF Configuration	CO3	T	Link-state concept, areas, OSPF terminology; Single-area OSPF setup, neighbor relationships	Lecture + Discussion	-

#	Session Topic	CO	L/T/P	Sub-Topics	Pedagogy	Assessment
20	OSPF Advanced Concepts & EIGRP Basics	CO3	T	Multi-area OSPF, DR/BDR election, cost calculation; Hybrid protocol concepts, feasible successor, metrics	Lecture + Discussion	-
21	EIGRP Configuration & Introduction to BGP	CO3	T	Lab-based configuration, verification commands; Inter-domain routing, Autonomous Systems, eBGP vs iBGP	Lecture + Discussion	-
22	BGP Concepts & Attributes	CO3	T	AS-path, next-hop, local preference, MED, path selection process	Lecture + Discussion	-
23	BGP Configuration & Lab	CO5	P	Configuring eBGP neighbors, verifying routes, basic troubleshooting	Lab Demonstration + Hands-on Practice	Lab Evaluation (BGP Configuration)
24	Access Control Lists (ACLs)	CO3	T	Standard vs extended ACLs, applying ACLs on routers	Lecture + Discussion	-
25	NAT (Network Address Translation)	CO3	P	Static NAT, dynamic NAT, PAT, use cases	Lab Demonstration + Hands-on Practice	-
26	Routing Lab & Final Review	CO5	P	Configure static + OSPF/EIGRP/BGP routing, troubleshooting scenarios, final quiz/exam	Lab Demonstration + Hands-on Practice	Quiz 3 + Routing Troubleshooting Assessment
27	Introduction to Network Security & Common Attack Types	CO4	T	CIA triad, common threats, attack surface, security terminology; Sniffing, spoofing (ARP/IP), MITM, DoS/DDoS, phishing overview	Lecture + Discussion	-
28	Firewalls & VPNs	CO4	T	Firewall types (packet-filtering, stateful, NGFW), VPN concepts, IPSec basics	Lecture + Discussion	-
29	Network Hardening Practices	CO4	T	Secure device configuration, AAA, SSH vs Telnet, wireless security (WPA2/WPA3)	Lecture + Discussion	-
30	Security Lab & Final Review	CO4	P	Configure ACL/firewall rules for basic defense, Wireshark traffic inspection, final assessment	Lab Demonstration + Hands-on Practice	Final Lab Evaluation + Final Assessment